



CERTIFICATE OF PRESENTATION

This certificate is awarded to

Lijia Zhang

for presenting the paper

"Beyond Scaling: A Stage 3 Geometric Framework for LLM Transparency through Language Manifold Dynamics"

at the Explainability and Transparency in AI (XTAI-2026) Symposium, held as part of the AISB Convention 2026 on 1 July 2026.

A handwritten signature in black ink, which appears to read 'Bertie Müller', is positioned above a horizontal line.

Dr Bertie Müller (AISB Chair)

Five Roads Ahead: Takeaways from the AISB 2026 Convention

Last week, I attended the AISB 2026 Annual Convention and presented my research on [language manifold dynamics](#) within the Explainability and Transparency in AI (XTAI) subgroup.

But the more lasting gift came from two sessions I attended outside my own track: Foundations of Computation in the morning, and Consciousness, Internal Language, and Chain-of-Thought Reasoning in the afternoon.

The ideas from those rooms did not stay separate. They reacted with questions already in my mind and left me with a clearer picture of where AI development might actually go. I came away seeing five distinct roads ahead: not competing roads, but branching paths.

1. The Scaling-Up Road

The theory behind this road is supported by an interesting phenomenon in physics: phase transition.

Water becomes steam. Laminar flow tips into turbulence. A slender column suddenly buckles under compression. In each case, gradual quantitative change eventually reaches a critical point and produces a qualitative transformation.

By analogy, the scaling-up road argues that artificial intelligence may follow a similar pattern. As the number of parameters, layers, training data, and computing power continue to increase, a neural network may eventually reach a critical mass where new forms of intelligence emerge.

This is the central bet: if we scale current AI systems by another major order of magnitude, perhaps 100 times the current computing power, artificial general intelligence may appear.

This road has already produced remarkable results. Whether it can truly produce general intelligence remains the open question.

2. The World-Model Road

Yann LeCun's road starts from a different premise: language is a fossil record of what human beings have already learned, encoded, and transmitted. The boundary of the world we know is the boundary of our language. But the real world is an iceberg, and our knowledge of it is only the visible tip.

In this sense, the world-model road is a move beyond epistemology toward ontology. Instead of focusing only on what humans have already known, named, and recorded in language, it asks AI to model what exists in the world itself.

LeCun may be right, but his road is likely to be long and winding. Starting from images, videos, and visual recognition, AI must eventually climb over the mountains of kinematics, dynamics, continuum mechanics, chemistry, biology, and the social sciences before entering the uncharted waters of physical reality itself.

3. The Agentic and Vertical Road

This is perhaps the most immediately productive direction. Rather than building one model that solves everything, this approach breaks a project into tasks, assigns them to the right tools or models, and uses an agent layer to coordinate the whole. Intelligence here is not located only inside the model. It emerges from the model's ability to orchestrate external capabilities.

Recent agent frameworks such as OpenClaw have demonstrated surprisingly strong early capability. By connecting AI models with files, tools, communication channels, and execution environments, they show how an LLM can move from answering questions to actually coordinating work.

With massive data collected across a complete production supply chain, China appears to be focusing strongly on vertical AI development, including discipline-specific and industry-specific AI. If successful, these specialized systems could be commanded, coordinated, and integrated by agentic AI, potentially forming a practical path toward AGI.

4. The Road of Fundamental Understanding

This road focuses on a deeper understanding of several basic concepts and their relationships: consciousness, perception, intelligence, and language. Instead of asking only how to make AI more powerful, it asks what intelligence actually is and how it arises.

This is where my own research sits. The work I presented at XTAI on geometric frameworks for LLM transparency is one small step along this road, an attempt to give language models a more physical and mathematical description.

Another subgroup approached these related questions from medical, neuroscience, and philosophical perspectives. Together, these discussions suggest that the future of AI may require not only larger models, but also deeper conceptual understanding.

5. The Evolutionary Road

No force on Earth is more powerful than evolution. It bends mountains, eliminates dinosaurs, creates intelligence from jellyfish, and may now be creating silicon intelligence from human civilization itself.

The conditions evolution requires are surprisingly minimal: enough variation, a selective regulatory environment, and time. Artificial evolution could potentially compress millions of years into meaningful engineering timescales through computation, simulation, and repeated selection.

But this road demands the most caution. An accelerated evolutionary process can generate outcomes beyond human prediction. If the selection criteria are poorly designed or the regulatory environment is unstable, the system may evolve toward goals that are efficient but unsafe, intelligent but misaligned. This brings AI safety, security, and governance from a side topic into a structural requirement.

These five roads are not in competition. They will converge, interact, and integrate.

AISB provides an important platform for this kind of cross-disciplinary collaboration. I hope AISB can maintain an active communication platform or online forum between conventions, so these conversations can continue beyond the annual event.

The lessons we learn from understanding, building, explaining, and governing AI may also spill over into other areas of academia, industry, and community life. In that sense, the journey toward AGI may carry the same significance as the goal of AGI itself.

[LLM Manifold Dynamics: XTAI 2026 Slides](#)

Full paper/preprint: <https://zenodo.org/records/20735385>

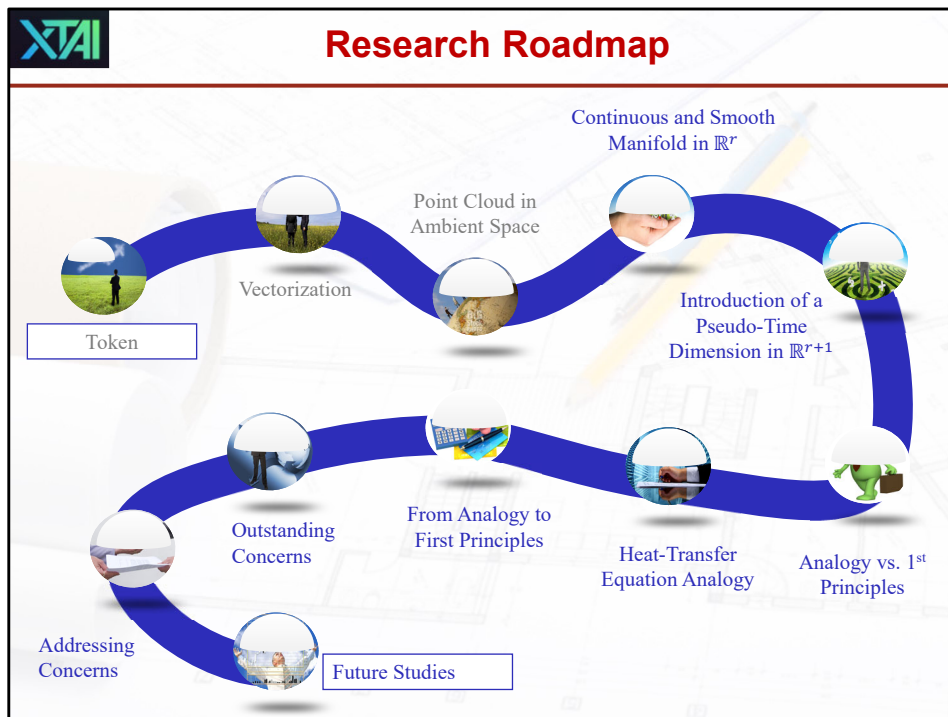


LLM in Manifold Expression

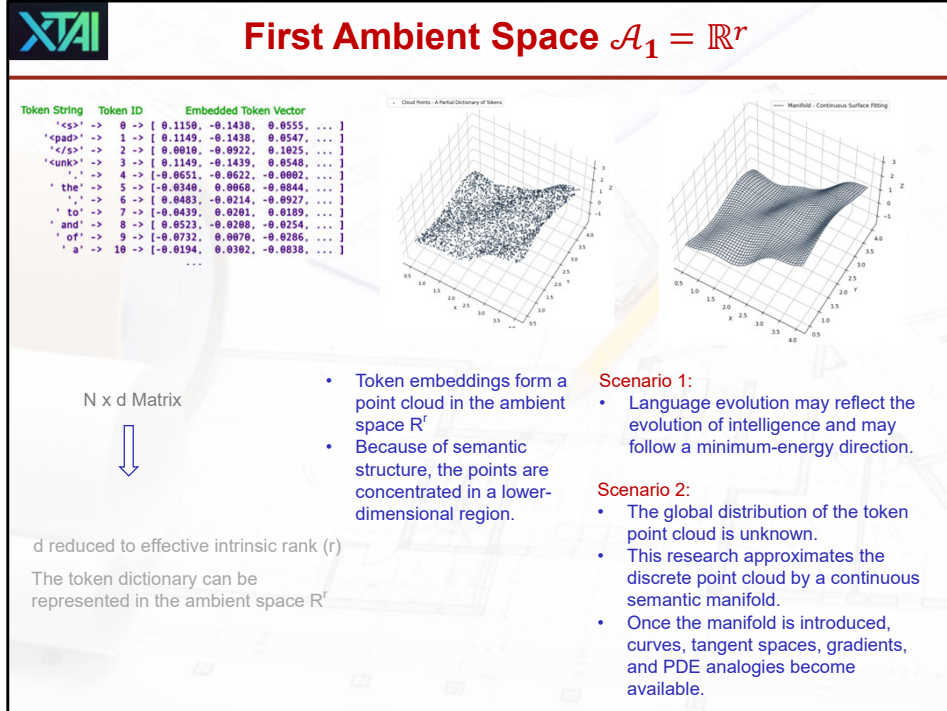
**Beyond Scaling: A Stage 3 Geometric Framework for LLM
Transparency through Language Manifold Dynamics**

7/1/2026

Lijia Zhang



Before I get into the details, let me first give you a roadmap for this research. During the presentation, I will occasionally return to this roadmap so that we can keep track of where we are in the overall argument.



Mathematically, the key step in this research is to move from discrete token embeddings to a continuous and smooth geometric structure. Once the discrete point cloud is approximated by a differentiable manifold, we can start using tools from differential geometry. We can define tangent spaces, gradients, scalar fields, and eventually partial differential equations, or PDEs. This is where my research begins: not from token vectorization itself, but from replacing the discrete point cloud with a continuous semantic manifold.



Second Ambient Space $\mathcal{A}_2 = \mathbb{R}^{r+1}$

Let $l=1,2,\dots,n$ denote the position of a token in a sequence of n tokens.

$\{1, \dots, n\} \times \mathbb{R}^r \implies \mathcal{A}_2 = \mathbb{R}^{r+1}$

- Replace the discrete token sequence with a continuous semantic curve.



Dynamics

$$\frac{\partial \Phi}{\partial t} = \kappa \nabla^2 \Phi$$

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \nabla^2 \Phi$$



Minkowsk-like Space

I like time.

When we are given a series of discrete items, the tools that immediately come to mind are usually combinatorics, permutations, or probability.

But physicists like time, because time gives us dynamics. The world around us is full of dynamic phenomena: motion, change, propagation, and evolution.

So once I replace the discrete token index with a continuous time-like parameter, I can start thinking about language dynamically. A token sequence is no longer just a list of isolated points. It can be approximated as a curve moving through a higher-dimensional semantic space.

For visual intuition, Minkowski-like space provides a useful analogy. It treats time as an additional dimension. Similarly, by introducing a pseudo-time dimension, the token sequence can be viewed as a trajectory in the second ambient space,

$$[\mathcal{A}_2 = \mathbb{R}^{r+1}.]$$

This is where PDE analogies become natural. Once the sequence is represented as a continuous curve with a time-like parameter, we can begin to discuss dynamics, diffusion, wave propagation, transport, and semantic evolution.



Analogy

Physical Analogy

$$\frac{\partial \Phi}{\partial t} = \kappa \nabla^2 \Phi$$

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \nabla^2 \Phi \qquad \frac{\partial^2 \Phi}{\partial t^2} + 2\beta \frac{\partial \Phi}{\partial t} = c^2 \nabla^2 \Phi$$

$$\frac{\partial \Phi}{\partial t} + u \cdot \nabla \Phi = 0$$

$$\frac{\partial^2 \Phi}{\partial t^2} + 2\beta \frac{\partial \Phi}{\partial t} + u(t) \cdot \nabla^2 \Phi = \nabla \cdot (K \nabla \Phi) + S(t)$$

Analogy gives candidate equations;
first principles gives meaning to the terms.

Back to First Principles

$$\nabla^2 \Phi = \nabla \cdot \nabla \Phi$$

?

What does the time derivative of
semantic potential actually mean?

Now let us try a dynamic formulation.

A dynamic problem is often described by a partial differential equation. In many PDEs, the time-dependent terms appear on one side, while the spatial-variation terms appear on the other side. This gives us a natural way to describe change, propagation, diffusion, transport, and evolution.

Before choosing a specific PDE, however, I want to step back and explain the thinking strategy behind this research.

There are two useful strategies. The first is analogy. The second is a first-principles approach.

I started from analogy because analogy gives us an immediate intuition before we go too deeply into the technical details. Once a time-like parameter is introduced, equations such as the heat equation, the wave equation, the transport equation, and diffusion-type equations naturally come to mind.

But analogy also has a limitation. Two systems may look similar mathematically while being very different in meaning. That is why we need to return to first principles and ask: what does each term really mean in the LLM setting? What does time mean?

What is the field? What is the space? What does the gradient represent?

So the method of this paper is to start from physical analogy, and then test whether the analogy can be justified through a first-principles interpretation.



Heat Transfer (Diffusion)

Heat Transfer (Diffusion)

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T$$

$$\alpha = \frac{k}{\rho c_p}$$

ρ Density of the material

c_p Specific heat capacity

k Thermal conductivity of the material

α Diffusivity coefficient

Analogy for LLM

$$\frac{\partial \Phi}{\partial t} = \kappa \nabla^2 \Phi$$

$$\frac{\partial \Phi}{\partial t} = \kappa \nabla_M^2 \Phi$$

Φ semantic potential

κ semantic diffusivity

∇_M^2 Laplace–Beltrami operator on M

The first PDE analogy I considered was the heat-transfer equation.

In classical heat transfer, temperature T evolves over space and time. The equation says that the rate of temperature changes depends on the spatial diffusion of temperature. The coefficient α , called thermal diffusivity, is defined by thermal conductivity divided by density and specific heat capacity.

The important point for this research is not the physical temperature itself, but the mathematical structure: a scalar field evolves over space through a diffusion operator. By analogy, I introduce a scalar semantic potential Φ . Instead of temperature diffusing through physical space, Φ represents a semantic field evolving over semantic space. However, the semantic space is not necessarily a flat Euclidean space. If the latent semantic space is modeled as a manifold, the ordinary Laplacian ∇^2 should be replaced by the Laplace-Beltrami operator ∇_M^2 .

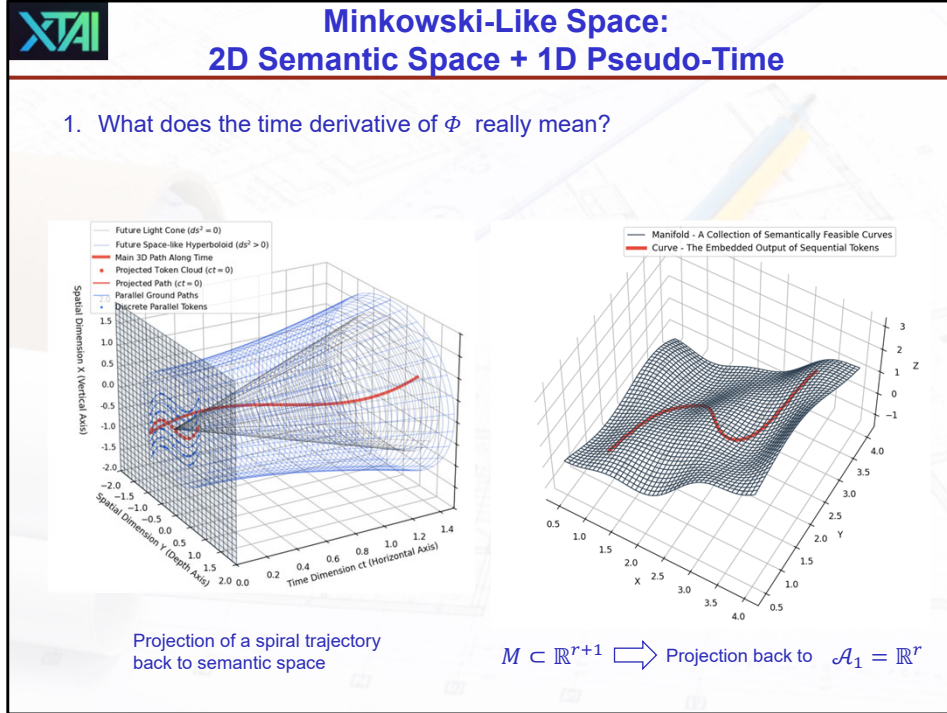
So the analogy begins with the heat equation, but the geometric version becomes semantic diffusion on a manifold.



First Principles Verification

1. What does the time derivative of Φ really mean?
2. What is the space, and what is the field?
3. How is the scalar field Φ tied to the manifold location p ?
4. On a semantic manifold, the Euclidean Laplace operator ∇^2 should be replaced by the Laplace-Beltrami operator Δ_M .

After introducing the heat-equation analogy, I need to step back and verify whether the analogy is meaningful from first principles. The key questions are: What does the time derivative of Φ mean in a token sequence? What is the space, and what is the field? How is Φ connected to a manifold location p ? And finally, if the semantic space is modeled as a manifold, the ordinary Laplacian should be replaced by the Laplace–Beltrami operator.



I use a 2+1 dimensional Minkowski-like space as an analogy to explain the significance of the time derivative in token space.

In the original semantic space, a token sequence appears only as an ordered set of points. We can project the trajectory back onto the semantic space to recover the token order, but this projection mostly tells us where the tokens are, not how fast the semantic state is moving.

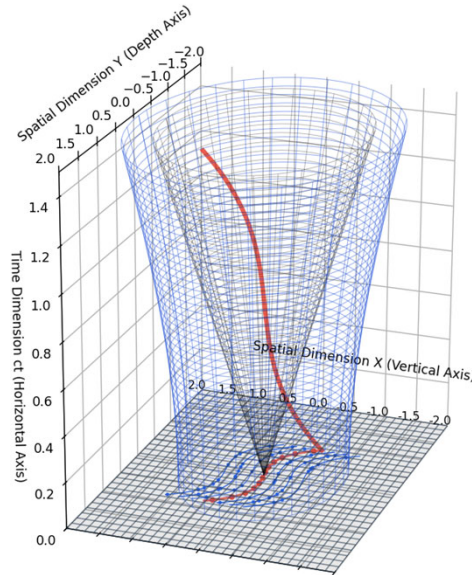
By introducing a pseudo-time dimension, the token sequence becomes a parameterized curve in a higher-dimensional space. The projection of this curve back to the semantic space gives the token order, while the time parameter gives us the rate of change along the sequence.

This rate of change may not be necessary if we only want to identify the token order. However, it becomes essential if we want to write a dynamic equation. PDEs require derivatives. Therefore, the pseudo-time derivative is introduced not simply to order the tokens, but to make semantic evolution differentiable and mathematically solvable.



Projection of a Parameterized Curve onto Semantic Space

1. What does the time derivative of ϕ really mean?



$\Gamma(t) = (ct, p(t))$, where

$p(t) \in M$, is the projected semantic trajectory on the semantic manifold.

$\Gamma(t)$ is the full curve in a Minkowski-like space.

The projection is:

$$\pi(\Gamma(t)) = p(t)$$

- Projection gives the token order and semantic path.
- Pseudo-time gives the rate of change.
- PDE modeling needs this differentiable parameter.

This slide explains why I introduce a pseudo-time dimension.

In the original semantic space, a token sequence can be viewed as an ordered set of points. If we only project the curve back to the semantic space, we can recover the token order, but we lose the rate at which the semantic state moves along the sequence.

By introducing a pseudo-time dimension, the token sequence becomes a parameterized curve in a higher-dimensional Minkowski-like space:

$$\Gamma(t) = (ct, p(t))$$

where $p(t)$ is the projected semantic trajectory on the semantic manifold, and c is a scaling factor for the pseudo-time dimension.

The projection is:

$$\pi(\Gamma(t)) = p(t)$$

The projected curve tells us the order and location of semantic movement, but PDE modeling requires more than order. It requires a differentiable parameter, so that semantic change can be expressed through derivatives such as:

$$\frac{dp}{dt}$$

or

$$\frac{\partial \Phi}{\partial t}$$

Therefore, the pseudo-time dimension is not introduced merely to reorder tokens. It is introduced to make semantic evolution differentiable and suitable for dynamic equation solving.



Connecting Φ Back to the Point p

The connection between Φ and p is functional dependence:

$$[\Phi = \Phi(p, t), \quad p \in M]$$

At each point p , the field has a scalar value:

$$[\Phi(p, t) \in \mathbb{R}]$$

The manifold gradient of Φ is a tangent vector at (p):

$$[\nabla_M \Phi(p, t) \in T_p M]$$

This tangent vector can guide the semantic trajectory:

$$\left[\frac{dp}{dt} = \pm \nabla_M \Phi(p, t) \right]$$

Therefore, Φ is connected to p not by equality, but by evaluation and differentiation on the manifold.

After clarifying the difference between space and field, the next question is how to connect the scalar potential Φ back to a specific semantic location p .

The key point is that Φ is not equal to p . It is also not equal to the token vector. Instead, Φ is a scalar field defined on the semantic manifold.

So the connection between Φ and p is functional dependence:

$$\Phi = \Phi(p, t), \quad p \in M$$

This means that at every semantic location (p), the field assigns a scalar value:

$$\Phi(p, t) \in \mathbb{R}$$

Once this is defined, the manifold gradient of Φ becomes meaningful:

$$\nabla_M \Phi(p, t) \in T_p M$$

This gradient is a tangent vector at p . It gives a local semantic direction on the manifold.

Therefore, Φ is connected to p in two steps: first by evaluation, $\Phi(p, t)$, and then by differentiation, $\nabla_M \Phi(p, t)$.

This tangent vector can then guide the semantic trajectory:

$$\frac{dp}{dt} = \pm \nabla_M \Phi(p, t)$$

The sign depends on whether the trajectory follows gradient ascent or gradient descent.



How to Define Potential Φ

2. What is the space, and what is the field?

3. How is the scalar field Φ connected to the manifold location p ?

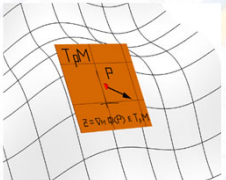
Heat-transfer analogy

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T$$

In heat transfer, T is a scalar field defined over a given Euclidean space.

$$\xi = \nabla_M \Phi(p) \in T_p M$$

First LLM prototype



$$\frac{\partial \Phi}{\partial t} = \kappa \nabla^2 \Phi$$

This direct analogy is not sufficient because:

- The semantic space is not necessarily Euclidean.
- Φ must be connected to a manifold location p .
- $\nabla \Phi$ in ambient space is not necessarily tangent to the manifold.

This slide explains why the first heat-transfer analogy was not sufficient by itself. In heat transfer, the space is usually given as Euclidean space, and temperature T is simply a scalar field defined over that space. But in the LLM case, the semantic space itself is part of the problem. It may be a curved manifold, not a neutral Euclidean background.

So I cannot simply replace temperature T with semantic potential Φ . I first need to explain what Φ is, where it is defined, and how it is connected to a point p on the semantic manifold.

The corrected formulation is to define (Φ) as a scalar field evaluated at a manifold location: $\Phi = \Phi(p, t)$, where $p \in M$. Then the manifold gradient $\nabla_M \Phi(p, t)$ becomes a tangent vector in $T_p M$, and this tangent vector can guide the semantic trajectory.



Ambient Gradient and Tangent Projection

3. How is the scalar field Φ connected to the manifold location p ?

First, define the semantic potential in the ambient embedding space:

$$\tilde{\Phi}: \mathbb{R}^r \rightarrow \mathbb{R}$$

Then restrict it to the semantic manifold:

$$\Phi = \tilde{\Phi}|_M$$

The ambient gradient is:

$$\nabla_{\mathbb{R}^r} \tilde{\Phi}(p) \left(\frac{\partial \tilde{\Phi}}{\partial z_1}, \frac{\partial \tilde{\Phi}}{\partial z_2}, \dots, \frac{\partial \tilde{\Phi}}{\partial z_r} \right)$$

But this gradient is not necessarily tangent to the semantic manifold:

$$\nabla_{\mathbb{R}^r} \tilde{\Phi}(p) \notin T_p M$$

Therefore, use the tangent projection:

$$\nabla_M \Phi(p) P_{T_p M} \left(\nabla_{\mathbb{R}^r} \tilde{\Phi}(p) \right)$$

Then the semantic trajectory is:

$$\frac{dp}{dt} \\ \nabla_M \Phi(p, t)$$

I first define the potential field in the ambient embedding space and then restrict it to the semantic manifold. The ambient gradient may contain components that are not tangent to the manifold. Since valid semantic movement should remain on the manifold, I project the ambient gradient onto the tangent space $T_p M$. This projected component becomes the manifold gradient, and it is this tangent vector that drives the semantic trajectory.



Euclidean Space → Minkowski-Like Space → Curved Semantic Manifold

4. On a semantic manifold, the Euclidean Laplace operator ∇^2 should be replaced by the Laplace–Beltrami operator ∇_M^2

Minkowski-Like Space

- Flat spacetime
- Adds a time-like dimension
- Useful for modeling trajectories

Riemannian Riemannian / Manifold Geometry

- Globally curved space
- Locally approximated by tangent spaces
- Requires manifold-based differentiation

Einstein's Spacetime Model

- Curved spacetime
- Space and time are unified geometrically
- Curvature changes the differential operators



$$ds^2 = dx^2 + dy^2 + dz^2 - c^2 dt^2$$

$$\square = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2$$

$$\nabla_g^2 \phi = \frac{1}{\sqrt{|g|}} \frac{\partial}{\partial x^i} \left(\sqrt{|g|} g^{ij} \frac{\partial \phi}{\partial x^j} \right)$$

$$\square_g \phi = \frac{1}{\sqrt{|g|}} \partial_\mu \left(\sqrt{|g|} g^{\mu\nu} \partial_\nu \phi \right)$$

where:
 $\mu, \nu = 0, 1, 2, 3$
and $x^0 = ct$ or $x^0 = t$.

This slide explains why the ordinary Euclidean Laplacian is not enough once the semantic space is modeled as a manifold.

In Euclidean space, the Laplace operator ∇^2 is defined with respect to a flat background. Minkowski space adds a time-like dimension and gives us a useful way to think about trajectories and dynamics.

But in this research, the semantic space is not assumed to be flat. It is better understood as a curved manifold. On a curved manifold, differentiation must respect the local geometry. That is why the ordinary Laplacian should be replaced by the Laplace-Beltrami operator.

Einstein's theory gives a useful analogy: once space and time are treated geometrically, curvature changes the form of the differential operators. Similarly, once semantic space is treated as a manifold, the PDE must be written using manifold-based operators.



Future Studies

Topological Continuity of Language

- The current framework assumes that the language manifold (M) is connected and smooth.
- This assumption may hold locally, but not globally.

Observation

- Within a topic, domain, or sentence, token transitions may form a locally smooth semantic curve.
- However, across unrelated domains, language may contain semantic jumps.
- These jumps may not be captured by a single connected manifold.

Future Direction

- A more general formulation may treat language space as a piecewise or disconnected structure:

$$M = M_1 \sqcup M_2 \sqcup \dots \sqcup M_k$$

where each M_i is a locally smooth semantic region.

- Tools from topology and algebraic topology, such as connected components, homology, and homotopy, may be needed to characterize the global semantic structure.

Beyond Scaling: A Stage 3 Geometric Framework for LLM Transparency through Language Manifold Dynamics

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April 2026

Abstract

This paper proposes a Stage 3 theoretical framework for understanding large language models (LLMs) through geometry and mathematical physics. Starting from a vocabulary embedding matrix $E \in \mathbb{R}^{N \times d}$, the paper identifies an intrinsic token semantic space $\mathbb{R}^r$, where r represents the effective semantic rank of the embedding representation. By adding the token sequence dimension as a temporal coordinate, we create a pseudo-time dimension; as such, the first ambient space is extended to a temporal-semantic ambient space $\mathbb{R}^{r+1}$. Observed language is then treated as discrete token samples or trajectories approximated, at first order, by a language manifold $M \subset \mathbb{R}^{r+1}$.

A scalar semantic potential Φ is introduced on the language manifold; its manifold gradient defines a tangent vector $\xi = \nabla_M \Phi$ describing the local direction and rate of steepest semantic change. The manifold and ambient space form a fixed geometric domain; Φ and its gradient are Eulerian fields defined over that domain, while a generated sequence, a *linguistic worldline*, is a Lagrangian trajectory traced through it by integrating a gradient-flow equation. In this formulation, the r -dimensional semantic space is analogized to a spatial field, while the token sequence dimension is treated as a pseudo-time domain. A token sequence can therefore be expressed as an ordered point cloud or trajectory embedded in the ambient space $\mathbb{R}^{r+1}$. However, language with semantic meaning tends to concentrate in smaller regions of this ambient space, which can be approximated by continuous manifolds. The diffusion equation provides a natural first candidate for fitting continuous manifolds to discrete linguistic samples, while wave and transport equations capture semantic propagation, structure preservation, and directional movement under contextual constraints. Together, these equations form a PDE-based framework for modeling language dynamics on the language manifold.

Training is interpreted as an inverse problem: estimating the language manifold, the scalar potential structure, and the coefficient fields of the governing PDE from human-generated language. Inference is interpreted as the forward problem: a prompt imposes boundary or initial conditions and selects a continuation trajectory on the learned manifold. The framework offers a path from statistical pattern recognition toward a predictive theory of language dynamics grounded in manifold geometry and PDEs.

1. Introduction: The Interpretability Crisis and the Missing Law

The dominant direction in LLM development, increasing model scale and improving benchmark performance, has delivered remarkable engineering achievements. Yet a persistent explanatory gap remains: LLM modeling remains largely biomimetic, attempting to emulate brain-like neural architecture and neural processing without fully uncovering the physical mechanisms that govern intelligence itself.

This situation is characteristic of a Stage 2 science. Following an epistemological ladder of

scientific progress, a discipline may be viewed as passing through four stages:

1. **Stage 1.** Raw observation and data collection.
2. **Stage 2.** Phenomenological pattern discovery, including scaling laws, empirical regularities, and benchmark correlations.
3. **Stage 3.** Identification of a governing theoretical framework that explains Stage 2 patterns as consequences.
4. **Stage 4.** Full mathematical formalization with predictive and quantitative power.

As an analogy, Galileo’s observations corresponded to Stage 1, while Kepler’s three empirical laws represented Stage 2 phenomenological pattern discovery in planetary motion. Newtonian mechanics then provided the Stage 3 theoretical framework and ultimately closed the knowledge loop by deriving those empirical patterns from first principles. Much of current LLM research remains largely in Stage 2.

The present paper first constructs two ambient spaces. The first is an intrinsic semantic space, analogized to a spatial domain. The second extends this semantic space by adding the token sequence dimension as a pseudo-time coordinate, thereby forming a temporal-semantic ambient space. Establishing this temporal-semantic space makes it possible to draw on existing tools from mathematical physics, where PDEs connect evolution in the time domain with variation in the spatial domain. In this framework, the governing PDE constrains the fitted language manifold, or family of manifolds, and its solutions represent admissible language trajectories or outputs.

2. Historical Lineage: From Energy Landscapes to Governing Laws

The present framework belongs to the long tradition of energy-based thinking in artificial intelligence. Boltzmann’s statistical mechanics introduced the idea that macroscopic states may emerge from microscopic configurations organized by an energy landscape. Hopfield networks later translated this idea into artificial cognition by representing stored patterns as attractor basins of a discrete dynamical system (Hopfield, 1982). The Boltzmann machine extended this view through stochastic learning and probability-weighted energy states (Hinton and Sejnowski, 1986). LeCun’s energy-based models further generalized the framework by assigning a scalar energy $E(x, y)$ to input–output pairs and interpreting inference as energy minimization (LeCun et al., 2006).

These approaches establish an important lineage: cognition can be understood through landscapes, attractors, and minimization. However, they do not fully specify the governing differential law that determines how a system evolves on the landscape. In other words, they provide energy functions or learning algorithms, but not the PDE that connects local changes in semantic potential, sequence evolution, and manifold geometry. This missing governing law is the Stage 3 gap addressed in this paper.

3. The Geometric Stack: From Vocabulary to Ambient Space

The framework is built in two sequential steps. The first step operates at the vocabulary level and constructs an intrinsic semantic space for individual tokens. The second step operates at the sequence level and appends a temporal coordinate corresponding to token order. Together, these steps define the mathematical arena in which language can be modeled as a manifold and, eventually, as a field governed by a partial differential equation. The stack may be summarized as follows:

$$E \in \mathbb{R}^{N \times d} \longrightarrow v_k \in \mathbb{R}^r \longrightarrow (i, v(i)) \in \mathbb{R}^{r+1} \longrightarrow M \subset \mathbb{R}^{r+1} \longrightarrow \text{prompt as boundary conditions.}$$

The first arrow is a vocabulary-level reduction that removes redundant embedding coordinates; the second arrow is a sequence-level extension that adds time.

3.1. Step 1: The Token Semantic Space $\mathbb{R}^r$

Following standard embedding practice, a language model represents its vocabulary by an embedding matrix

$$E \in \mathbb{R}^{N \times d},$$

where N is the number of tokens and d is the model-imposed latent embedding dimension. The dimensionality of word embeddings has been studied theoretically by [Yin and Shen \(2018\)](#), who show that dimensionality selection can be analysed as a mathematical problem rather than treated as an arbitrary model parameter choice.

Each row of E represents one token as a vector

$$e_k \in \mathbb{R}^d, \quad k = 1, \dots, N.$$

The vocabulary embedding matrix therefore represents N discrete token embeddings as a finite point cloud in the continuous latent space $\mathbb{R}^d$:

$$\{e_k\}_{k=1}^N \subset \mathbb{R}^d.$$

Although each token embedding lives in $\mathbb{R}^d$, the finite set of vocabulary embeddings may span only an r -dimensional subspace of $\mathbb{R}^d$, where

$$r = \text{rank}(E), \quad r \leq \min(N, d).$$

Equivalently, $\text{span}\{e_1, \dots, e_N\} \cong \mathbb{R}^r$. Here r denotes the intrinsic linear semantic dimension of the vocabulary representation. This rank-based view is consistent with empirical work by [Kataiwa et al. \(2025\)](#), who find that token embedding spaces often have much lower intrinsic dimensionality than their extrinsic embedding dimensions. We therefore define the first ambient space as

$$\mathcal{A}_1 = \mathbb{R}^r.$$

3.2. Step 2: The Temporal-Semantic Ambient Space $\mathbb{R}^{r+1}$

If the intrinsic token semantic space $\mathbb{R}^r$ is interpreted as an r -dimensional spatial-like vector space, then a token sequence can be viewed as the evolution of a semantic state along an additional temporal dimension. The sequence index plays the role of time, while the r semantic coordinates play the role of spatial dimensions.

This discrete-to-continuous interpretation is consistent with the Neural ODE framework of [Chen et al. \(2018\)](#), who show that discrete neural-network transformations can be interpreted as an Euler discretisation of continuous dynamics. In their formulation, the hidden state evolves according to $dh/dt = f_\theta(h, t)$, rather than through a fixed finite sequence of layers. Although the present paper studies token trajectories on a semantic manifold rather than hidden-layer dynamics, the same mathematical idea motivates treating a discrete sequence index as a continuous temporal variable.

Let $i = 1, 2, \dots, n$ denote the token position in a sequence. At each position i , the token is represented by a semantic vector $v(i) \in \mathbb{R}^r$. Each contextualised token is therefore a pair $(i, v(i))$ lying in the product space $\{1, \dots, n\} \times \mathbb{R}^r$. In the continuum approximation the discrete index i may be treated as a temporal variable t , and the product space is approximated by the temporal-semantic ambient space

$$\mathcal{A}_2 = \mathbb{R}^{r+1}.$$

Thus the second ambient space is obtained by appending one temporal dimension to the intrinsic token semantic space: $\mathbb{R}^r \rightarrow \mathbb{R}^{r+1}$. We write $\mathcal{A} \equiv \mathcal{A}_2 = \mathbb{R}^{r+1}$ for the ambient space used throughout the remainder of the paper.

In this formulation, language can be understood as a set of discrete observed points or trajectories lying on, or near, a continuous manifold embedded in $\mathbb{R}^{r+1}$. The discrete tokens are the sampled observations, while the underlying language manifold represents the continuous geometric structure that constrains admissible linguistic sequences. We denote this manifold by

$$M \subset \mathbb{R}^{r+1}.$$

This manifold contains the admissible trajectories of language: sequences that are grammatically coherent, semantically meaningful, and pragmatically interpretable within a linguistic community. Language is not designed from first principles. It evolves through repeated human use, social selection, communicative efficiency, memory constraints, and cultural transmission. Over time, this evolutionary process causes admissible linguistic patterns to converge toward stable structures in the temporal-semantic ambient space. In the terminology of this paper, these stable structures define a basin of attraction, which we identify as the language manifold M . The manifold is not created by a large language model; rather, it is gradually shaped by the historical evolution of human language and preserved in the fossil record of human-generated texts.

The observed tokens and token sequences are discrete samples in $\mathbb{R}^{r+1}$. Collectively, they form a point cloud in the temporal-semantic space $\mathcal{A}$. The language manifold $M \subset \mathbb{R}^{r+1}$ is the continuous lower-dimensional structure fitted to, or inferred from, this point cloud. The role of the governing PDE is to provide a smooth constraint on this fit, so that admissible linguistic trajectories are not arbitrary curves through the ambient space, but paths lying on or near M .

It is useful to distinguish M itself from any single trajectory lying on it. The manifold M represents the entire admissible region, the full space of sequences a linguistic community would accept. A single generated token sequence, by contrast, is one particular curve traced through M . Borrowing terminology from physics, where a particle’s path through spacetime is called its worldline, we refer to such a curve as a *linguistic worldline*: the specific trajectory realized by one act of language production, as opposed to the manifold of all trajectories that could have been produced.

Section 4 turns to the governing law: what type of partial differential equation governs evolution, diffusion, propagation, or collapse on M ?

4. Governing PDE on the Language Manifold

When a manifold is embedded in a Euclidean or Riemannian space, it may be viewed as a lower-dimensional geometric structure within the ambient space. A flat plane or affine subspace can be described by linear equations, while a curved manifold often requires differential operators to characterise its geometry. In this sense, a PDE provides a general mathematical framework for representing, constraining, or recovering the shape of a manifold. In the language setting, observed tokens or token sequences can be understood as discrete sample points concentrated in a region of the temporal-semantic ambient space. A governing PDE may then be interpreted as a smooth, PDE-constrained fit to this cloud of discrete linguistic samples. Among possible candidates, the diffusion equation is especially useful because diffusion geometry shows that the Laplacian associated with heat flow can encode and recover the geometry of an underlying manifold from sampled data (Jones, 2024). Recent work on implicit manifold-valued diffusions further supports this view by constructing diffusion processes on data manifolds from point-cloud samples without requiring explicit charts or projections (Kawasaki-Borruat et al., 2026). Therefore, a diffusion-type equation provides a natural first candidate for modeling the language

manifold, while wave and transport equations may capture additional effects such as semantic propagation and directional flow.

4.1. Building the Diffusion Equation

Let $v \in \mathbb{R}^r$ denote the intrinsic semantic vector of a token. By adjoining the sequence coordinate t , the token is represented as a point

$$p = (v, t) \in M \subset \mathbb{R}^{r+1}$$

on the language manifold. Let $\Phi : M \rightarrow \mathbb{R}$ denote a scalar semantic potential, analogous to temperature T in heat transfer. At each point $p \in M$, the manifold gradient of Φ defines a tangent vector

$$\xi = \nabla_M \Phi(p) \in T_p M,$$

where $T_p M$ is the tangent space of the language manifold at p . Thus, v represents the token’s intrinsic semantic coordinate on the manifold, while the intrinsic gradient $\xi \in T_p M$ defines a local tangent direction field encoding the orientation and rate of steepest semantic increase. It characterizes the manifold’s local heading at p , not a literal chordal displacement between adjacent discrete tokens in the ambient space. Because $p = (v, t)$, the tangent space $T_p M$ is itself $(r + 1)$ -dimensional, so ξ in general has both a semantic component and a sequence-coordinate component. Writing $\pi_{\text{sem}}(v, t) = v$ for the projection onto the intrinsic semantic coordinates, we denote this semantic component $\xi_{\text{sem}} = \pi_{\text{sem}}(\xi) = \pi_{\text{sem}}(\nabla_M \Phi)$. It is this projected vector, not the full tangent vector ξ , whose negative gives the worldline’s velocity in the gradient-flow equation below.

This setup establishes a dichotomy between the field-wide and trajectory-wise characterizations of language. The ambient space $\mathbb{R}^{r+1}$ and the language manifold $M \subset \mathbb{R}^{r+1}$ are the fixed geometric reference domain; they are not themselves Eulerian or Lagrangian, but the arena in which both descriptions are stated. Within that arena, the scalar potential Φ and its gradient $\nabla_M \Phi$ are Eulerian objects: fields defined over the entire manifold M at once, governed by the PDE developed below. A linguistic worldline (Section 3.2), by contrast, is a Lagrangian object, a single moving point

$$\gamma(s) = p(s) = (v(s), t(s)) \in M,$$

not a field. Under the natural parametrization $t(s) = s$, the worldline reduces to $\gamma(s) = (v(s), s)$, and its semantic component evolves by the projected gradient-flow equation

$$\frac{dv}{ds} = -\xi_{\text{sem}}(\gamma(s)) = -\pi_{\text{sem}}(\nabla_M \Phi(\gamma(s))).$$

Only the semantic component of ξ drives v ; the sequence coordinate advances independently, at the fixed unit rate fixed by the parametrization $t(s) = s$, rather than being driven by the t -component of ξ . In this view, language modeling decomposes into two sequential steps: first solve the field-wide problem to recover Φ over M (training, Section 5.1); second, given a fixed point $v(0)$, integrate the gradient-flow equation forward to obtain the worldline (inference, Section 5.2). Only the worldline, not the entire solved field, corresponds to the observable generated sequence.

This separation also clarifies a potential objection: if Φ is solved over the whole manifold M , including positions already realized by earlier tokens, in what sense is the field dynamic during generation? It is not. Under this formulation, Φ is solved once, as a boundary- or initial-value problem over the whole of M , and does not change as generation proceeds; sequence position is internalized as a coordinate of M rather than as a clock driving a time-varying field. All of the dynamics visible during text generation are Lagrangian: they belong entirely to the worldline $\gamma(s)$ as it moves through the already-fixed field Φ , not to Φ itself.

The term “worldline” is borrowed from physics by analogy only. The language manifold is equipped with a Riemannian, positive-definite metric, not a Lorentzian one, and the framework introduces no causal light-cone structure or propagation-speed constant. Where a bound on the rate of semantic change is needed, it is more naturally stated as a constraint on the gradient norm, $\|\xi\| \leq \xi_{\max}$, than as a literal speed limit.

In the heat-transfer analogy, the rate of change of the scalar potential in the time domain is proportional to the Laplacian of that potential in the spatial domain. Therefore, in the local flat approximation, the semantic diffusion equation is

$$\frac{\partial \Phi}{\partial t} = \kappa \nabla^2 \Phi,$$

where $\kappa > 0$ is the semantic diffusion coefficient. In the heat-transfer analogy, κ corresponds to the thermal diffusivity of the medium, which controls how rapidly temperature changes in time in response to spatial temperature curvature. Here, the “time domain” corresponds to the token sequence dimension, while the “spatial domain” corresponds to the intrinsic semantic directions on the language manifold. Consistent with Φ being solved once, field-wide, $\partial \Phi / \partial t$ denotes the derivative of that field with respect to the sequence coordinate t , not evolution in any external physical time; t is a coordinate of M , not a clock. Throughout this subsection, ∇ , ∇^2 , ∇_M , ∇_M^2 , K , and div_M act only on the intrinsic semantic directions, with t held as the separate coordinate against which these spatial operators evolve; none of them differentiate with respect to t itself.

This steady-state reading is sharpened by distinguishing t from a second, external parameter τ , the actual generation/evolution time at which the field-solving step (Section 5.1) is regarded as complete and frozen. With this distinction, the steady-state condition is

$$\frac{\partial \Phi(p, \tau)}{\partial \tau} = 0 \quad \text{for every fixed } p \in M,$$

i.e. once solved, Φ does not change at any manifold point as generation proceeds. This is compatible with Φ being spatially nonuniform over M , including along the sequence coordinate t that is already built into $p = (v, t)$: in general $\Phi(p_1) \neq \Phi(p_2)$ for $p_1 \neq p_2$, and $\partial \Phi / \partial t$, the variation of the solved field along the sequence coordinate, need not vanish. The two statements address different questions and do not conflict: $\partial \Phi / \partial \tau = 0$ says the field itself is fixed once solved, while $\partial \Phi / \partial t \neq 0$ says that fixed field is not flat along the sequence direction. A worldline accordingly experiences a changing potential as it moves, purely because it visits different points of an already-fixed, spatially varying field,

$$\frac{d}{ds} \Phi(\gamma(s)) = g(\nabla_M \Phi(\gamma(s)), \dot{\gamma}(s)),$$

which is generically nonzero even though $\partial \Phi / \partial \tau = 0$ throughout. In short: the field is steady in the external sense (τ -invariant) but not uniform in the spatial/sequence sense (t -varying); the worldline’s experience of change is entirely Lagrangian, arising from its motion through that fixed-but-nonuniform field, consistent with the Eulerian/Lagrangian decomposition introduced above.

The ordinary Euclidean Laplacian ∇^2 should be understood as a local flat approximation. If the language manifold is globally curved, ∇^2 should be replaced by the Laplace–Beltrami operator ∇_M^2 . In a standard three-dimensional heat-transfer problem, the Euclidean Laplacian ∇^2 is adequate because heat diffuses through a Euclidean domain. However, if heat transfer is constrained to a curved thin shell or surface, the diffusion domain is itself a curved manifold, and the Laplace–Beltrami operator is required to account for curvature. By analogy, language dynamics on a curved language manifold should be written as

$$\frac{\partial \Phi}{\partial t} = \kappa \nabla_M^2 \Phi.$$

Here ∇_M^2 incorporates the Riemannian metric of the language manifold. In local coordinates, this introduces metric-dependent terms, equivalently expressed through covariant derivatives or Christoffel symbols.

In the language manifold, semantic diffusion is unlikely to be isotropic. The intrinsic semantic directions are not independent, and activation does not spread equally in every direction. Some dimensions may be strongly coupled, such as subject, tense, agency, modality, causality, and discourse role, while others may interact only weakly. The scalar diffusion coefficient κ should therefore be replaced by a semantic conductivity tensor K . In the local flat approximation the anisotropic diffusion equation becomes

$$\frac{\partial \Phi}{\partial t} = \nabla \cdot (K \nabla \Phi).$$

The diagonal entries of K describe diffusion along individual semantic directions; the off-diagonal entries describe coupling between different semantic dimensions. If both curvature and anisotropy are considered, the most general diffusion form becomes

$$\frac{\partial \Phi}{\partial t} = \text{div}_M(K \nabla_M \Phi),$$

which is the natural generalization of heat diffusion to an anisotropic semantic field on a curved language manifold.

4.2. General PDE Forms for Geometry Fitting

The diffusion equation provides a useful geometric fitting tool because it connects time-domain evolution with spatial-domain curvature and attraction toward stable structures. However, language dynamics are unlikely to be purely diffusive, like water cascading downhill. Language does not merely smooth or dissipate semantic activation. It also preserves structure, exhibits resonance, propagates long-range dependencies, and moves directionally under contextual constraints.

For this reason, the wave equation may be more consistent with certain features of the language manifold, especially where semantic structures persist and propagate across long sequences. Similarly, transport equations may be needed to describe directional semantic movement under the influence of context. Therefore, rather than relying on a single diffusion equation, this paper considers a broader family of PDE-based geometric fitting models.

The common feature of this PDE family is that each equation connects evolution in the time domain with structure in the spatial domain. In the language setting, the time domain corresponds to the token sequence dimension, while the spatial domain corresponds to intrinsic semantic directions on the language manifold. These equations therefore provide different ways to represent token flow following the gradient structure of semantic potential.

4.3. Wave Equation: Semantic Propagation and Structure Preservation

To model the preservation and propagation of linguistic structure, we introduce the wave equation. In the flat approximation:

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \nabla^2 \Phi.$$

On the curved language manifold, this becomes

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \nabla_M^2 \Phi,$$

where c represents the propagation velocity of semantic structure.

A more general form includes a damping term, which provides additional flexibility for fitting semantic dynamics:

$$\frac{\partial^2 \Phi}{\partial t^2} + 2\beta \frac{\partial \Phi}{\partial t} = c^2 \nabla_M^2 \Phi,$$

where $\beta > 0$ is the damping coefficient. This term allows temporary semantic effects to decay while stable discourse structures continue to propagate. Thus, the wave equation may fit the part of language dynamics involving semantic memory, structure preservation, and long-range dependency.

4.4. Transport Equation: Directional Semantic Flow

A third candidate is the transport equation. In the flat approximation,

$$\frac{\partial \Phi}{\partial t} + u \cdot \nabla \Phi = 0,$$

where u is a semantic velocity field tangent to the language manifold. The vector field u is different from both the intrinsic semantic vector v and the gradient tangent vector $\xi = \nabla_M \Phi$.

While v represents the intrinsic semantic coordinate of a token and ξ the local direction and rate of steepest semantic change induced by the potential Φ , u represents the direction and speed of semantic transport along the manifold. This distinction gives the model additional flexibility: token flow does not have to follow only the gradient of the token’s intrinsic attributes. It may also be guided by contextual direction, discourse intention, or prompt-imposed constraints. The gradient-flow equation of Section 4.1 is the special case $u = -\xi_{\text{sem}}$, the semantic projection of ξ introduced there, in which transport follows the potential’s own gradient rather than an independently specified direction.

On a curved manifold, the transport equation becomes

$$\frac{\partial \Phi}{\partial t} + u \cdot \nabla_M \Phi = 0,$$

where u lies on the tangent space of M and $\nabla_M \Phi$ is the intrinsic gradient on the manifold.

4.5. Toward a General Governing PDE

The three PDE forms can be combined into a more general governing template:

$$\frac{\partial^2 \Phi}{\partial t^2} + 2\beta \frac{\partial \Phi}{\partial t} + u(t) \cdot \nabla_M \Phi = \text{div}_M(K \nabla_M \Phi) + S(t). \quad (1)$$

In this equation:

- $\partial^2 \Phi / \partial t^2$ represents wave-like propagation,
- $2\beta \partial \Phi / \partial t$ represents damping or dissipation,
- $u(t) \cdot \nabla_M \Phi$ represents transport along a semantic flow field,
- $\text{div}_M(K \nabla_M \Phi)$ represents anisotropic diffusion on the curved language manifold, and
- $S(t)$ represents source terms introduced by new tokens, external context, or multimodal input.

This general PDE is not proposed as the final equation of language. Rather, it is a governing-law template: it shows how language dynamics may combine smoothing, propagation, and directional flow on the language manifold.

5. Training and Prompting as Inverse and Forward PDE Problems

5.1. Training as an Inverse PDE Problem

The governing PDE of the language manifold is not known in advance. Neither the exact form of the equation nor its coefficient fields are directly given. Therefore, LLM training may be interpreted as an inverse PDE problem. In the terms of Section 4.1, training solves the Eulerian field problem, recovering Φ over the whole manifold M , rather than tracing any single worldline.

The fossil record of human-generated language provides discrete samples of trajectories lying on, or near, the language manifold M . Language is not constructed from an architectural blueprint. It evolves through repeated human use, social selection, communicative efficiency, memory constraints, and cultural transmission. Over time, admissible linguistic patterns converge toward stable structures in the temporal-semantic ambient space. In the terminology of this paper, these stable structures define an attractor basin, which we identify as the language manifold M .

The language manifold is embedded in the second ambient space, $\mathcal{A} \equiv \mathcal{A}_2 = \mathbb{R}^{r+1}$, which is a linear temporal-semantic space. From this perspective, LLM training is not the construction of language from raw data, but the search for, and approximation of, a pre-existing language manifold within $\mathcal{A}$. The learned model estimates the geometry of M from discrete linguistic samples.

Training also estimates the scalar potential structure Φ and the coefficient fields of the governing PDE, including the semantic conductivity tensor $K(t)$, the transport vector field $u(t)$, the damping coefficient β , and the source term $S(t)$. These quantities are not prescribed in advance. They are inferred from the observed distribution of human-generated language.

In this interpretation, model weights are not merely memorised statistical associations. They are distributed approximations of the geometry and governing dynamics of the language manifold. The model learns how semantic activation diffuses, propagates, and transports across M under different contextual and boundary conditions.

5.2. Prompting as Boundary Condition, Slicing, and Collapse

Inference is the corresponding forward problem. Given a prompt P , the model imposes boundary or initial conditions on the language manifold M . These constraints produce a prompt-conditioned scalar potential field Φ_P . In the terms of Section 4.1, inference is the Lagrangian step: given the already-solved field Φ_P and a fixed initial point, it integrates the gradient-flow equation forward to trace the resulting worldline.

The prompt is therefore not merely an input string. It plays a central role in reducing the admissible region of the language manifold. In the temporal-semantic formulation, the ambient space $\mathcal{A} = \mathbb{R}^{r+1}$ is a linear coordinate space, while the language manifold $M \subset \mathbb{R}^{r+1}$ is a lower-dimensional Riemannian manifold embedded within it. The prompt supplies a finite set of token positions and semantic states, and therefore fixes the part of the trajectory that any admissible continuation must satisfy.

This reduction can be understood in two complementary ways. From a linear-algebra perspective, the prompt imposes a system of constraints on the full temporal-semantic ambient space. The query-key-value mechanism supplies contextual relationships that restrict the admissible degrees of freedom, similar to using equations to reduce the dimensionality of a solution space. From a PDE perspective, the prompt supplies boundary or initial conditions for the governing equation. The model response is then the continuation trajectory selected under these constraints.

In transformer architectures, this constraint is operationalised through the query-key-value attention mechanism. The prompt tokens generate queries, keys, and values that determine

which prior semantic states are relevant to the next state. Geometrically, this process can be interpreted in three related ways. First, the prompt imposes boundary values by fixing known points on the manifold. Second, it slices the manifold by assigning values to certain contextual or semantic dimensions, thereby reducing the admissible solution set to a lower-dimensional submanifold. Third, through attention-weighted aggregation, it may collapse or average information across selected token positions, producing a lower-dimensional effective representation of the context.

Thus, prompting is a geometric operation on the language manifold. By imposing boundary conditions, fixing parameters, slicing admissible dimensions, and aggregating relevant information, the prompt progressively reduces the full language manifold to a constrained admissible region. The output of a language model can therefore be interpreted as a trajectory selected from this reduced region of M . The response is a continuation on the language manifold under constraints supplied by the prompt.

The admissible continuation may also be interpreted as a stable or low-energy path on the manifold, analogous to a physical system evolving toward an attractor or minimum-free-energy configuration.

The relationship between training and inference can therefore be summarized as follows. Training estimates the geometry of M , the scalar potential structure Φ , and the coefficient fields governing semantic diffusion, propagation, and transport. Prompting then imposes boundary or initial conditions on this learned structure, reducing the admissible region of the manifold. The response is the continuation trajectory selected from this constrained region.

6. Discussion, Limitations, and Future Research

The framework proposed in this paper is supported by several related lines of existing work.

First, the reduction from the model-imposed latent dimension d to the intrinsic semantic rank r is motivated by recent work on the intrinsic dimensionality of token embeddings. [Kataiwa et al. \(2025\)](#) suggest that token embeddings contain substantial redundancy and may lie in a lower-dimensional intrinsic representation space. This supports the first step of the present framework: replacing the full embedding space $\mathbb{R}^d$ with an intrinsic token semantic space $\mathbb{R}^r$, where r captures the effective number of independent semantic directions.

Second, diffusion geometry provides a mathematical foundation for fitting continuous geometric structures to discrete point-cloud samples. [Jones \(2024\)](#) shows that diffusion operators, Laplacians, and carré-du-champ structures can recover geometric information from sampled data. [Kawasaki-Borruat et al. \(2026\)](#) further develop diffusion processes on implicit manifolds, supporting the idea that a manifold can be inferred from point-cloud data through diffusion-based operators. This directly supports the interpretation of observed token sequences as discrete samples lying on, or near, a continuous language manifold M .

Third, Neural ODEs provide a precedent for introducing a time dimension into neural computation. [Chen et al. \(2018\)](#) interpret discrete neural-network updates as an Euler discretisation of continuous dynamics and replace a fixed sequence of hidden layers with an ordinary differential equation for the hidden state. This supports the second step of the present framework: treating the discrete token index as a continuous pseudo-time variable t . The time dimension is the missing piece that connects the semantic spatial domain $\mathbb{R}^r$ with sequence evolution, producing the temporal-semantic ambient space $\mathbb{R}^{r+1}$. Once this space is constructed, PDEs become natural candidates for governing the relationship between time-domain evolution and spatial-domain semantic structure.

Fourth, work on data-driven discovery of intrinsic dynamics provides a useful precedent for

recovering lower-dimensional dynamics from high-dimensional observations (Floryan and Graham, 2022). This is relevant to the present framework because the assumption of a continuous language manifold should ultimately be treated as an approximation rather than as a completed geometric characterization. In high-dimensional embedding space, lower-dimensional language structure may appear as a smooth manifold, but it may also appear as a cluster structure, graph-like structure, stratified space, singular manifold-like set, or other non-smooth geometric object. Therefore, the language manifold proposed in this paper should be understood as a first smooth approximation to the underlying semantic structure.

Fifth, the proposed language-manifold framework is consistent with a broader two-ambient-space approach to representation learning. In a related formulation for image-based world models, Zhang (2026) introduces a two-ambient-space continuum reformulation of I-JEPA, in which the image-plane chart domain is connected with the patch-content domain through an advection-diffusion PDE. The present paper applies the same general principle to LLMs: an intrinsic semantic ambient space is first constructed and then extended by a pseudo-time coordinate, allowing PDE-based tools to connect sequence evolution with semantic variation.

In the present framework, the term “initial state” has two related but distinct meanings. First, the pretrained language manifold may be regarded as an unactivated background semantic landscape, denoted M_0 , whose learned geometry already contains latent gradients and admissible directions of semantic evolution. Second, once a prompt or first token is introduced, it defines an activated initial point or state $q_0 \in M_0$. The subsequent states then evolve from q_0 along the learned semantic landscape, forming a trajectory whose observable trace corresponds to the generated language output. This trajectory is precisely the linguistic worldline of Section 3.2, obtained by integrating the gradient-flow equation of Section 4.1 forward from q_0 through the field Φ already fixed by training on M_0 .

These lines of work correspond to the major steps of this paper:

$$\begin{aligned} \text{intrinsic rank reduction} &\longrightarrow \mathbb{R}^r, \\ \text{addition of pseudo-time dimension} &\longrightarrow \mathbb{R}^{r+1}, \\ \text{diffusion/PDE fitting} &\longrightarrow M \subset \mathbb{R}^{r+1}. \end{aligned}$$

The current proposal should be understood as a Stage 3 framework rather than a completed predictive theory. Future work may further develop this framework toward Stage 4 by providing empirical validation, quantitative operators, and more rigorous geometric formulations. Several open problems remain: estimating the intrinsic semantic rank r ; recovering the semantic potential Φ and its gradient field $\xi = \nabla_M \Phi$ empirically from data; replacing the local flat operator ∇^2 with the manifold operator ∇_M^2 ; identifying which PDE terms are required for language dynamics; and recovering coefficient fields such as K , u , β , and S from trained model activations.

A further direction is to develop an atlas-based formulation of language geometry. Instead of assuming one globally smooth language manifold, the global semantic structure may be covered by local charts or manifold-like patches M_α , each representing a local semantic, syntactic, pragmatic, or domain-specific region. Transition maps and compatibility conditions between neighboring charts would then be needed to determine whether these local structures form a smooth manifold, a stratified manifold, or a more general geometric object.

Another direction is to connect manifold identification with a fibre-bundle-like factorization of the embedding space. Identifying a lower-dimensional semantic structure and constructing such a factorization can be viewed as two complementary forms of dimensional reduction. The former seeks a lower-dimensional semantic substructure within the high-dimensional embedding space, while the latter decomposes the total embedding space into a task-relevant semantic base and auxiliary fibres that may represent style, context, modality, domain, or redundant degrees of

freedom. The projection from the total embedding space to a lower-dimensional base space can be interpreted as eliminating, collapsing, or marginalizing variations along the fibre. However, this separation is task-dependent: a dimension that is irrelevant for one task may be important for another. Therefore, a rigorous formulation would require a defined projection map or a learned procedure for determining which variations belong to the semantic base and which belong to the auxiliary fibres.

Finally, the rigorous formulation of PDE operators on the probability simplex is left for future study. Since LLM outputs are probability distributions over vocabulary tokens, a complete theory should eventually connect the temporal-semantic manifold formulation with operators defined on, or coupled to, the probability simplex. This would clarify how semantic potential fields, token probabilities, and manifold dynamics interact during inference.

Although the present paper focuses on LLMs, the framework may provide a bridge from language modeling to real-world AI. LLMs are the most accessible case because token embeddings and sequence positions naturally define a temporal-semantic ambient space. In industry-specific or discipline-specific AI, the same idea may be extended by replacing the language manifold with a domain manifold built from engineering, financial, legal, biological, or operational data. In this broader view, LLMs are not the endpoint, but the first workable example of a general framework in which intelligence is modeled as manifold learning plus governing dynamics. This provides a possible path from LLM transparency toward real-world AI, industry-specific AI, discipline-specific AI, and eventually AGI.

7. Conclusion

This paper has proposed a Stage 3 theoretical framework for understanding large language models through the geometry of language manifolds and the dynamics of partial differential equations. The two-step construction, from vocabulary embeddings to an intrinsic semantic space $\mathbb{R}^r$, and from there to a temporal-semantic ambient space $\mathbb{R}^{r+1}$, provides a principled arena in which language can be modeled as a continuous manifold fitted to discrete token trajectories. Training is interpreted as the inverse problem of recovering this manifold and its governing PDE from human-generated text; inference is interpreted as the corresponding forward problem, in which a prompt imposes boundary conditions and selects an admissible continuation trajectory.

The framework does not claim to be a completed quantitative theory. Rather, it offers a Stage 3 conceptual bridge between the empirical scaling laws of Stage 2 and the predictive, first-principles formalism of Stage 4. The open problems identified in the Discussion, intrinsic rank estimation, empirical recovery of the potential Φ and its gradient field $\xi = \nabla_M \Phi$, atlas-based geometry, fibre-bundle factorization, and PDE coefficient recovery from model activations, mark the path forward toward a fully quantitative theory of language dynamics.

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